

Week 5 — Session 3

Course: 10211CS225 — Problem Solving and Testing using Java

Unit II: Algorithmic Thinking & Competitive Problem Patterns

Topic: String Handling Basics

Language: Java

Focus: Practice parsing, character inspection, and efficient one-pass string processing.

Task 1: String to Integer (atoi)

Practice: <https://leetcode.com/problems/string-to-integer-atoi/>

Complete Java Solution

```
import java.util.*;

public class Main {
    static int myAtoi(String s){
        int i=0,n=s.length(),sign=1;
        while(i<n && s.charAt(i)==' ') i++;

        if(i<n && (s.charAt(i)=='+' || s.charAt(i)=='-')){
            if(s.charAt(i)=='-') sign=-1;
            i++;
        }

        long value=0;
        while(i<n && Character.isDigit(s.charAt(i))){
            value=value*10+(s.charAt(i)-'0');
            long signed=sign*value;
            if(signed>Integer.MAX_VALUE) return Integer.MAX_VALUE;
            if(signed<Integer.MIN_VALUE) return Integer.MIN_VALUE;
            i++;
        }
        return (int)(sign*value);
    }

    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        System.out.println(myAtoi(sc.nextLine()));
        sc.close();
    }
}
```

Task 2: Alternating Characters

Practice: <https://www.hackerrank.com/challenges/alternating-characters/>

Complete Java Solution

```
import java.util.*;

public class Main {
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int t=sc.nextInt();

        while(t-->0){
            String s=sc.next();
            int deletions=0;
            for(int i=1;i<s.length();i++)
                if(s.charAt(i)==s.charAt(i-1)) deletions++;
            System.out.println(deletions);
        }
        sc.close();
    }
}
```

Programs are standalone Java solutions. Java 17+ is recommended.